DISCRETE STRUCTURES

CMSC250

Chapter 23: Functions and Their Properties

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Mappings, injectivity, surjectivity, and inverses



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1 What is a function?

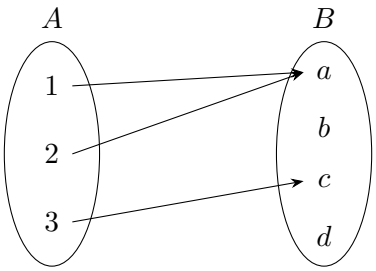
A *function* $f : A \to B$ assigns **exactly one** output $f(a) \in B$ to every input $a \in A$. The set A is the *domain*; B is the *codomain*; and the *range* (or image) is $f(A) = \{f(a) : a \in A\}$. The range need not equal the codomain.

Key idea. A function may send several inputs to the same output, but one input may not have two outputs. A function must also define an output for every element of its domain.

Worked example: domain, codomain, and range

Question. Let $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$, and $f(1) = a$, $f(2) = a$, $f(3) = c$. Identify the domain, codomain, and range.

Solution. The domain is A, the codomain is B, and the range is $\{a, c\}$. Notice that b, d are in the codomain but are not attained.



2 Injective functions (one-to-one)

A function $f : A \to B$ is *injective* if different inputs always produce different outputs. Formally,

$$\forall x, y \in A, \quad f(x) = f(y) \implies x = y.$$

Equivalently, $x \neq y \implies f(x) \neq f(y)$. The standard proof strategy is to assume $f(x) = f(y)$ for arbitrary domain elements and derive $x = y$.

Worked example: prove a linear function injective

Question. Prove $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = 3n + 2$, is injective.

Proof. Let $a, b \in \mathbb{Z}$ be arbitrary. Assume $f(a) = f(b)$. By the definition of f, $3a + 2 = 3b + 2$. Subtracting 2 and dividing by 3 yields $a = b$. Therefore f is injective. ■

Why the proof works. The proof begins with equal outputs, not equal inputs, because that is exactly the condition in the definition of injectivity.

Worked example: disprove injectivity

Question. Is $g : \mathbb{Z} \to \mathbb{Z}$, $g(n) = n^2$, injective?

Solution. No. The distinct inputs 1 and -1 satisfy $g(1) = g(-1) = 1$. One collision disproves injectivity.

3 Surjective functions (onto)

A function $f : A \to B$ is *surjective* if every element of the *codomain* is attained: $\forall b \in B\, \exists a \in A\, [f(a) = b]$. To prove surjectivity, start with an *arbitrary target* $b \in B$, solve $f(a) = b$ for a candidate input, and verify the candidate belongs to A.

Worked example: a surjective function

Question. Prove $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x - 5$, is surjective.

Proof. Let $y \in \mathbb{R}$ be arbitrary. Choose $x = (y + 5)/2$. Since $y \in \mathbb{R}$ and the real numbers are closed under addition and division by nonzero real numbers, $x \in \mathbb{R}$. Moreover, $f(x) = 2((y + 5)/2) - 5 = y$. Thus every y in the codomain has a preimage, so f is surjective. ■

Worked example: codomain changes the answer

Question. Compare $h : \mathbb{Z} \to \mathbb{Z}$, $h(n) = 2n$, with $k : \mathbb{Z} \to 2\mathbb{Z}$, $k(n) = 2n$.

Solution. The function h is not onto $\mathbb{Z}$ because, for example, 1 is not twice an integer. The function k is onto $2\mathbb{Z}$ because every even integer y has the form $2n$ for some integer n. The formula is the same, but the codomain changes surjectivity.

4 Bijective functions

A function is *bijective* if it is both injective and surjective. A bijection pairs each input with exactly one output and every output with exactly one input.

Worked example: prove bijectivity

Question. Prove $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = n + 7$, is a bijection.

Proof. For injectivity, let $a, b \in \mathbb{Z}$ and suppose $f(a) = f(b)$. Then $a + 7 = b + 7$, hence $a = b$. For surjectivity, let $y \in \mathbb{Z}$ be arbitrary and choose $x = y - 7 \in \mathbb{Z}$. Then $f(x) = y$. Therefore f is both injective and surjective, so it is bijective. ■

5 Inverse functions and inverse images

If $f : A \to B$ is bijective, its *inverse function* $f^{-1} : B \to A$ maps each output to its unique input. The defining identities are $f^{-1}(f(a)) = a$ for all $a \in A$ and $f(f^{-1}(b)) = b$ for all $b \in B$.

Worked example: find and check an inverse

Question. Find the inverse of $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x - 4$.

Solution. Write $y = 3x - 4$ and solve: $x = (y + 4)/3$. Thus $f^{-1}(y) = (y + 4)/3$. Checking, $f(f^{-1}(y)) = 3((y + 4)/3) - 4 = y$, and $f^{-1}(f(x)) = ((3x - 4) + 4)/3 = x$. For *any* function $f : A \to B$, the inverse *image* (preimage) of a subset $S \subseteq B$ is

$$f^{-1}(S) = \{a \in A : f(a) \in S\}.$$

This set is defined even if f is not bijective. It is not the same concept as an inverse function.

Worked example: inverse image without an inverse function

Question. For $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = n^2$, find $f^{-1}(\{1, 4\})$.

Solution. We need all integers whose squares are 1 or 4: $f^{-1}(\{1, 4\}) = \{-2, -1, 1, 2\}$.

6 Composition of functions

For $f : A \to B$ and $g : B \to C$, the composition $g \circ f : A \to C$ is defined by $(g \circ f)(x) = g(f(x))$. The right-hand function acts first.

Worked example: order of composition matters

Question. Let $f(x) = x + 2$ and $g(x) = x^2$ on $\mathbb{R}$. Compute $g \circ f$ and $f \circ g$.

Solution. $(g \circ f)(x) = (x + 2)^2$, whereas $(f \circ g)(x) = x^2 + 2$. In general these are different.

7 Images of sets

For $S \subseteq A$, the image is $f(S) = \{f(x) : x \in S\}$. A useful fact is that $f(S \cup T) = f(S) \cup f(T)$ for all $S, T \subseteq A$. **Question.** Prove $f(S \cup T) = f(S) \cup f(T)$.

Proof. Let y be arbitrary. Then $y \in f(S \cup T)$ iff there exists $x \in S \cup T$ with $f(x) = y$. This is equivalent to there being an $x \in S$ with $f(x) = y$ or an $x \in T$ with $f(x) = y$. That is equivalent to $y \in f(S)$ or $y \in f(T)$, hence to $y \in f(S) \cup f(T)$. Since membership is equivalent for every y, the sets are equal. ■

8 Practice questions with complete solutions

Question. 1. Is $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = n^3$, injective? Surjective?

Solution. Injective: if $a^3 = b^3$ then $a = b$ (cubing is strictly increasing). Not surjective: 2 is not an integer cube. **Question.** 2. Prove $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$, is bijective.

Proof. If $a^3 = b^3$ then $a = b$, so f is injective. Given any $y \in \mathbb{R}$, take $x = \sqrt[3]{y} \in \mathbb{R}$; then $f(x) = y$, so f is surjective. Thus f is bijective. ■**Question.** 3. For $f(n) = 2n + 1$ on $\mathbb{Z}$, find the preimage of $\{3, 5, 8\}$.

Solution. Solving $2n + 1 = 3$ gives $n = 1$ and solving $2n + 1 = 5$ gives $n = 2$. The equation $2n + 1 = 8$ gives $n = 7/2 \notin \mathbb{Z}$. Therefore the preimage is $\{1, 2\}$. **Question.** 4. If $f : A \to B$ and $g : B \to C$ are injective, prove $g \circ f$ is injective.

Proof. Let $a_1, a_2 \in A$ and assume $g(f(a_1)) = g(f(a_2))$. Since g is injective, $f(a_1) = f(a_2)$. Since f is injective, $a_1 = a_2$. Therefore $g \circ f$ is injective. ■

9 Common mistakes

Do not confuse codomain with range. Do not start an injectivity proof by assuming $a = b$; assume equal outputs. For surjectivity, verify that your constructed preimage lies in the stated domain. An inverse image of a set exists even for a non-bijective function.